

NEWCASTLE INTL, United Kingdom

WMO: 032433

Lat: 55.038N

Lon: 1.692W

Elev: 81

StdP: 100.35

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.2	(d) -6.2	(e) 2.3	(f) -2.1	(g) -5.0	(h) 2.5	(i) -0.9	(j) 16.8	(k) 8.3	(l) 14.5	(m) 7.8	(n) 1.3	(o) 280	(p) 0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	23.1	17.4	21.8	16.9	20.1	15.8	18.4	22.1	17.4	20.6	16.4	19.3	3.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.3	20.1	16.0	11.5	18.9	15.1	10.8	17.9	52.1	21.9	49.1	20.4	46.2	19.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.5	9.2	-7.2	26.7	2.3	1.5	-8.9	27.8	-10.2	28.6	-11.5	29.5	-13.2	30.5
			-7.2	20.6	2.4	1.2	-8.9	21.4	-10.3	22.1	-11.6	22.8	-13.4	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	4.0	4.4	5.6	7.6	10.3	13.0	15.1	15.1	12.9	9.9	6.4	4.2
	DBStd	4.94	2.98	3.04	2.98	2.86	2.91	2.51	2.40	2.26	2.38	2.76	3.05	3.55
	HDD10.0	931	187	157	138	83	32	3	0	0	3	36	112	181
	HDD18.3	3390	444	389	394	323	249	161	103	105	163	262	358	438
	CDD10.0	592	0	1	3	9	41	94	159	157	91	32	4	1
	CDD18.3	9	0	0	0	0	0	1	4	3	1	0	0	0
	CDH23.3	47	0	0	0	0	1	6	22	16	2	0	0	0
	CDH26.7	3	0	0	0	0	0	0	1	2	0	0	0	0
Wind	WSAvg	3.7	4.5	4.5	4.4	3.7	3.5	3.3	3.0	3.1	3.3	3.5	3.8	4.0
Precipitation	PrecAvg	698	55	46	42	52	50	62	63	66	57	66	77	65
	PrecMax	1036	146	98	97	146	106	169	153	163	141	137	166	144
	PrecMin	524	17	7	9	8	13	14	10	12	17	16	24	27
	PrecStd	115	29	25	25	41	25	40	33	31	33	27	35	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.8	15.0	18.9	22.1	24.1	25.9	25.8	23.1	18.2	15.0	13.2
		MCWB	10.5	10.9	10.7	13.5	16.5	17.6	18.4	18.9	17.1	14.9	13.1	11.3
	2%	DB	11.1	11.9	13.0	15.9	19.1	21.2	23.2	22.2	20.2	16.8	13.2	12.0
		MCWB	9.1	9.6	10.2	11.7	14.1	16.0	17.4	17.4	16.0	14.3	11.4	10.2
	5%	DB	9.9	10.2	11.9	14.0	17.1	19.8	21.8	20.9	18.2	15.2	12.1	10.9
		MCWB	8.2	8.2	9.2	10.4	12.8	15.2	17.0	16.8	14.6	13.2	10.4	9.2
	10%	DB	8.8	9.1	10.2	12.8	15.2	18.0	20.0	19.2	17.1	14.2	11.0	9.2
		MCWB	7.5	7.4	7.9	9.7	11.6	14.0	16.0	15.3	14.1	12.2	9.4	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	11.2	12.0	13.8	16.9	19.0	19.6	19.8	18.3	15.6	13.3	12.1
		MCDB	12.5	13.1	14.8	17.7	21.7	23.4	23.8	22.7	22.0	17.5	14.6	13.4
	2%	WB	9.5	9.9	10.6	12.1	15.0	16.8	18.4	18.3	16.6	14.6	12.0	10.6
		MCDB	10.9	11.6	12.6	15.2	18.5	20.1	22.1	21.7	19.5	16.3	13.3	11.9
	5%	WB	8.3	8.7	9.5	10.9	13.4	15.4	17.4	17.2	15.5	13.6	10.9	9.5
		MCDB	9.5	10.5	11.4	13.6	16.4	18.7	20.6	20.2	17.7	15.3	12.1	10.8
	10%	WB	7.2	7.5	8.4	9.9	12.2	14.4	16.3	16.2	14.6	12.6	9.8	8.2
		MCDB	8.2	8.9	10.3	12.2	15.1	17.5	19.3	18.6	16.7	14.2	10.7	9.4
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	6.5	7.4	7.9	7.7	8.0	7.9	7.4	6.4	5.4	4.9
		MCDBR	6.0	6.5	8.3	9.9	10.8	10.8	10.7	10.0	9.5	7.5	6.0	6.1
		MCWBR	5.4	5.4	5.9	6.4	6.8	6.6	6.0	5.9	6.0	5.8	5.3	5.6
	5% WB	MCDBR	5.8	6.3	7.5	9.2	10.1	9.8	9.6	8.8	8.4	7.0	5.8	6.1
		MCWBR	5.5	5.5	5.9	6.4	6.8	6.5	6.0	5.8	5.9	5.8	5.4	5.7
Clear-Sky Solar Irradiance	taub	0.316	0.333	0.364	0.387	0.385	0.387	0.388	0.378	0.358	0.342	0.316	0.299	
	taud	2.310	2.380	2.310	2.270	2.324	2.346	2.349	2.391	2.449	2.451	2.369	2.277	
	Ebn at Noon	611	747	802	833	858	859	850	836	806	729	610	538	
	Edh at Noon	56	74	98	117	117	116	114	103	85	68	52	46	
All-Sky Solar Radiation	RadAvg	0.56	1.20	2.33	3.79	4.92	4.92	4.71	3.90	2.77	1.51	0.71	0.42	
	RadStd	0.07	0.13	0.28	0.37	0.28	0.41	0.39	0.26	0.23	0.13	0.06	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page